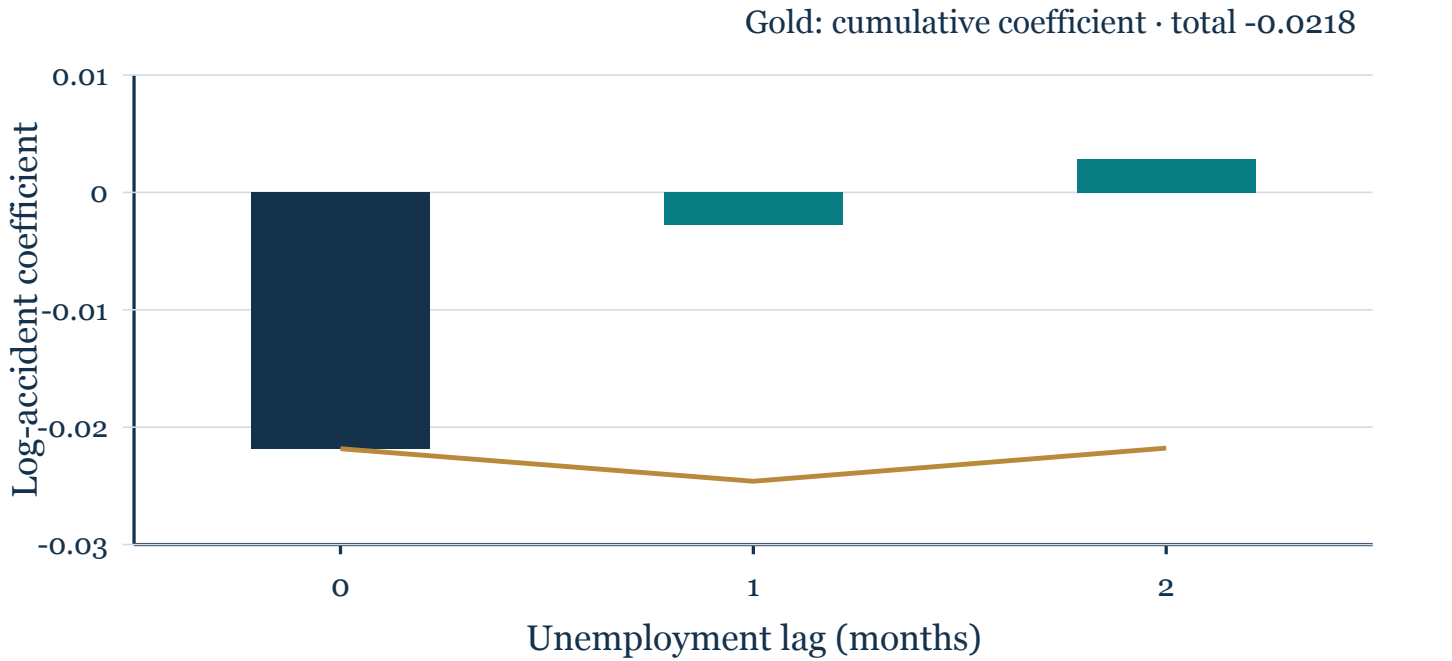
# Does unemployment affect accidents only this month?

Allow a relationship to unfold over several periods.




Actual monthly unemployment coefficients from the finite distributed-lag regression. Cumulative means the sum through the selected lag.

# Define lags, differences and horizons

They are different operations on ordered observations.

> $LX_t = X_{t-1}$

> $\Delta X_t = X_t - X_{t-1}$

# Write a finite distributed-lag model

Every coefficient has an outcome date and a regressor date.

$$Y_t = a + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \text{controls} + u_t$$

# Trace a one-period regressor change

Its outcome responses occur at several dates.

> Horizon $h$: $0, 1, 2, 3$

> Temporary-input responses: $0.20, 0.10, -0.05, 0$

> Cumulative response: $0.20 + 0.10 - 0.05 = 0.25$

# A permanent change has another time path

The long-run level effect reaches the lag sum in this model.

> Permanent-input responses: $0.20, 0.30, 0.25, 0.25, \ldots$

> Temporary cumulative response and permanent long-run level contrast are different quantities.

# Lag sum

Try the comparison before opening the explanation.

**Practice**

A finite distributed-lag model has coefficients 0.2, 0.1 and −0.05 on current x, lag 1 and lag 2. Compute the impact and the cumulative response to a one-period unit x shock.

# Lag sum: explanation

Follow the calculation and interpret its units.

> Impact is 0.2. The three-period cumulative response is $0.2 + 0.1 - 0.05 = 0.25$ outcome units. In this finite-lag model, a permanent unit x increase has a long-run level effect of 0.25 as well. An autoregressive dependent-variable model has a different multiplier formula.

# Introduce the uncertainty option before reading the lag table

Newey–West changes the covariance estimate around an unchanged OLS fit.

> Newey–West/HAC: allow suitable unequal spread and time dependence in uncertainty estimation.

> Four covariance lags here mean four months; the mean equation is unchanged.

# Read the observed lag estimates

Let the application determine the pattern.

```
. newey logacc unem L(1/2).unem trend i.month, lag(4)

------------------------------------------------------------------------------
             |             Newey–West
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        unem |
         --. |  -.0218345   .0073752    -2.96   0.004    -.0364866   -.0071823
         L1. |  -.0027711   .0075824    -0.37   0.716    -.0178349    .0122927
         L2. |   .0028242   .0079337     0.36   0.723    -.0129375    .0185858
             |
       trend |   .0018359   .0002514     7.30   0.000     .0013365    .0023352
             |
       _cons |   10.67207   .0367848   290.12   0.000     10.59899    10.74515
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

# Estimate uncertainty for the lag sum

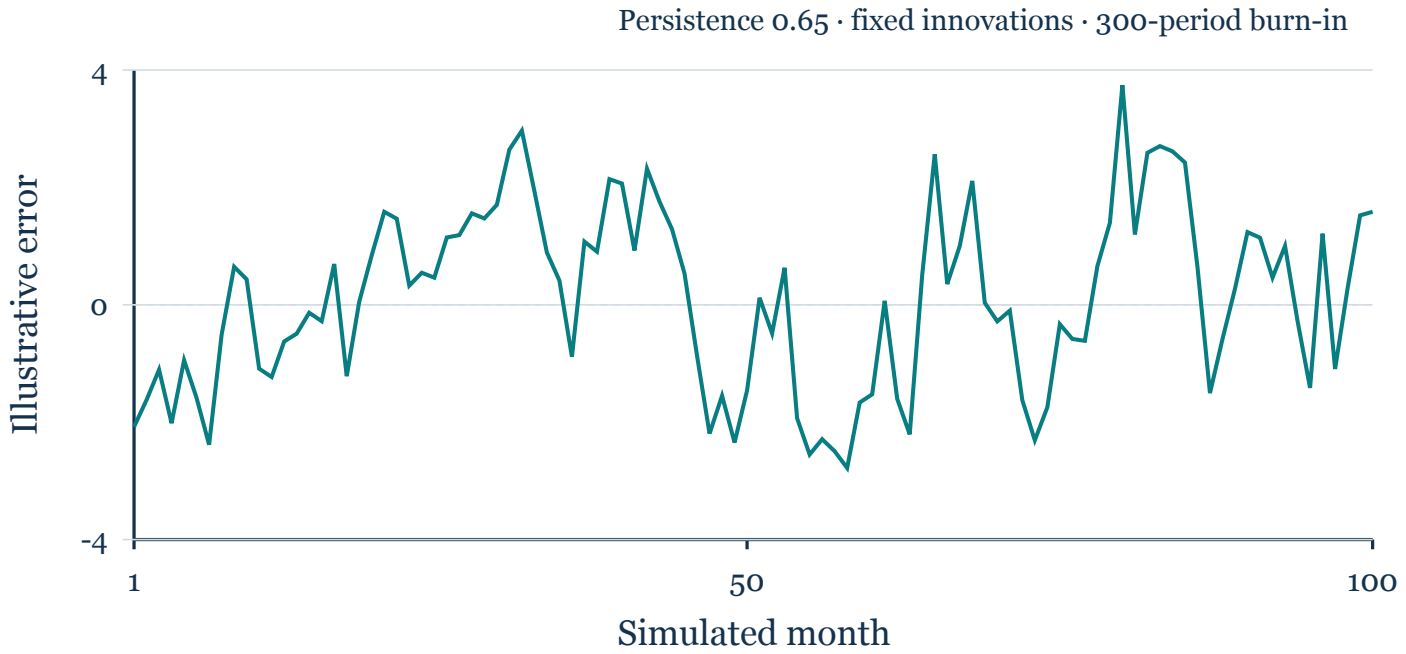
The coefficient covariances matter.

```
. lincom unem+L.unem+L2.unem

------------------------------------------------------------------------------
      logacc | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |  -.0217814    .004154    -5.24   0.000    -.0300341   -.0135287
------------------------------------------------------------------------------
```

# An AR(1) model carries the outcome forward

The persistence parameter determines whether shocks fade.




Illustrative simulation · fixed innovations · vertical range adapts to the selected persistence

$Y_t = a + \rho Y_{t-1} + \varepsilon_t$

One innovation with $\rho = 0.6$: $1, 0.6, 0.36, 0.216, \ldots$

# A unit root does not let shocks fade in the usual way

A random walk accumulates innovations.

> Random walk: $Y_t = Y_{t-1} + \varepsilon_t$

> If $Y_0 = 0$, $\mathrm{Var}(Y_t) = t\sigma^2$

> $\Delta Y_t = \varepsilon_t$

# Stationarity describes stable statistical features

Specify which definition is being used.

> Covariance stationary: stable finite mean/variance and lag-dependent covariance.

> AR(1) stationary regime requires $|\rho| < 1$ and suitable innovations/initialization.

> Strict stationarity concerns full joint distributions.

# A permanent input change can be amplified by outcome persistence

Solve for the steady difference instead of adding coefficients blindly.

> $\Delta Y_h = 0.6\Delta Y_{h-1} + 0.2: \quad 0.20, 0.32, 0.392, \ldots$

> $\Delta Y = 0.6\Delta Y + 0.2 \Rightarrow \Delta Y = 0.5$

> Long-run multiplier: $\beta/(1-\rho)$ for a stable recursion.

# AR long run

Try the comparison before opening the explanation.

**Practice**

For $Y_t = a + 0.6Y_{t-1} + 0.2X_t + e_t$, what is the long-run response to a permanent unit increase in x under a stable model?

# AR long run: explanation

Follow the calculation and interpret its units.

> The steady-state change solves $\Delta Y = 0.6\Delta Y + 0.2$, so $\Delta Y = 0.2/(1 - 0.6) = 0.5$. This calculation assumes a stable dynamic relation and an exogenous permanent x change with other determinants fixed.

# Inspect persistence in the accident model

The fitted lag is a conditional association.

```
. newey logacc L.logacc trend i.month, lag(4)

------------------------------------------------------------------------------
             |             Newey–West
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      logacc |
         L1. |   .6783359   .0825587     8.22   0.000     .5143906    .8422811
             |
       trend |   .0009218   .0002548     3.62   0.000     .0004158    .0014278
             |
       _cons |   3.276324   .8753799     3.74   0.000     1.537993    5.014655
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

# Differencing changes the target relationship

It can remove an accumulating level component in a specific model.

> For the simple random walk, first differencing leaves the innovation.

> A level relationship and a relationship between changes answer different empirical questions.

# Unrelated trending series can appear related

A high R-squared can be misleading.

> Two independent series with accumulating random shocks can produce a striking regression fit in levels even without an economic relationship.

> Inspect time plots and model the source of persistence before interpreting a slope.

# Trending regression

Try the comparison before opening the explanation.

**Practice**

Two unrelated series both rise over time and produce a high $R^2$ when regressed on each other. What should you do before interpreting the slope?

# Trending regression: explanation

Follow the calculation and interpret its units.

> Inspect their time plots and consider deterministic trends, stochastic trends and stationarity. A high $R^2$ does not establish a meaningful relationship. Choose a specification and inference procedure justified by the series' properties; differencing every series automatically is also not a substitute for reasoning.

# Weak dependence concerns information far apart in time

Persistence and large-sample reasoning must be compatible.

> Weak dependence: sufficiently distant information becomes weak enough for the stated limit result.

> Stationarity alone does not guarantee that time averages behave like independent draws.

# The sample changes when lags are added

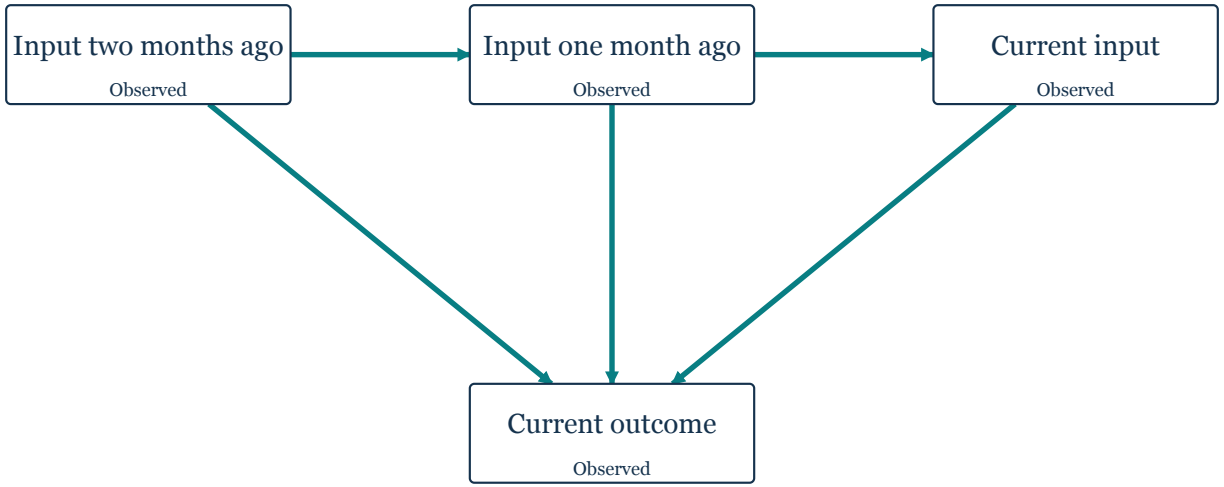
Compare fit and tests on compatible observations.

> A two-lag model loses the first two observations if no earlier data are available.

> Define the common estimation window before comparing models or applying a fit-loss test.

# Explain a dynamic result with a time path

One label such as effect is not enough.

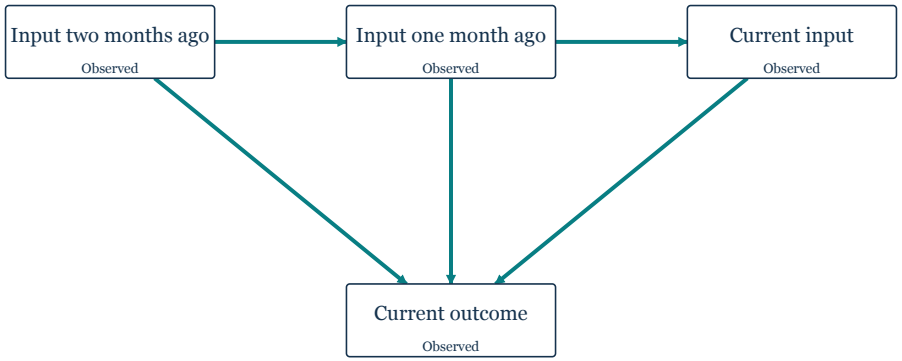



3 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

# DAG practice: partial lag effect or propagated input change?

Trace the routes before choosing a control.




3 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

## Practice

An earlier input influences the outcome directly and by changing later input values. Which routes remain if the later inputs are held fixed? Why can the response to a naturally propagated input innovation differ from the coefficient on that earlier input?

# DAG practice: partial lag effect or propagated input change?: explanation

Explain what conditioning changes in this stated graph.

> Holding both later inputs fixed blocks the routes through those input nodes. The direct earlier-input-to-current-outcome route remains. Its partial contrast differs from an intervention that also changes later values of the input process. For a propagated innovation, trace the resulting entire input path and combine it with the distributed-lag coefficients. For a one-period imposed regressor increase, specify that all other dates remain fixed. A sum of lag coefficients is meaningful for the stated path and model; the word shock alone does not define which path was changed.

# Trace the actual monthly lag pattern

Reproduce the evidence, then explain its economic meaning.

**Practice**

Run meeting_10.do. Record the current, one-month-lagged and two-month-lagged unemployment coefficients. Calculate their sum and compare it with lincom. Explain the difference between the sequence after a one-month unit increase and the long-run level contrast after a permanent increase in this finite-lag model.

# Trace the actual monthly lag pattern: explanation

A reproducible result needs both a calculation and an interpretation.

> The three coefficients are -0.021834, -0.002771 and 0.002824. Their sum is -0.021781 in log-accident units per unemployment-rate unit. The covariance among estimates matters for the SE of that sum, which is why lincom is useful. A one-month unit increase has those dated fitted responses, then no further direct effect in this finite two-lag specification. Their sum is the cumulative log-outcome response. A permanent unit increase eventually activates all three terms at once, giving the same sum as a long-run level contrast. An equation with a lagged outcome needs a different multiplier calculation and stability assumptions.

# Next, examine what the fitted errors still remember

Dynamics and uncertainty are related but distinct.

> We have defined stationary behaviour, distributed lags and autoregressive persistence, and distinguished several dynamic contrasts.

> The next lesson studies serial correlation in what the model leaves unexplained.